

## Question

### What is thermodynamics?

- A** Study of the relationship between heat and other forms of energy.
- B** Study of the conversion of chemical energy to other forms of energy.
- C** Study of the relationship between mechanical energy to other forms of energy.
- D** Study of the conversion of mechanical energy to other forms of energy.

## Question

**Statement-I: System is a part of universe.**

**Statement-II: System is a part of surrounding.**

- A** Statement I and Statement II both are correct.
- B** Statement I is correct but Statement II is incorrect.
- C** Statement I is incorrect but Statement II is correct.
- D** Statement I and Statement II both are incorrect.

## Question

**A system is said to be heterogeneous when it;**

- A** is completely uniform throughout.
- B** is consist of one phase only.
- C** is not uniform throughout.
- D** has uniform composition throughout.

## Question

**System that cannot exchange both energy and matter with the surroundings is called;**

- A** open system.
- B** closed system.
- C** isolated system.
- D** None of these.

## Question

Match List-I with List-II to find out the correct option.

| List-I |                 | List-II |                                    |
|--------|-----------------|---------|------------------------------------|
| A.     | Rigid wall      | I.      | Movable                            |
| B.     | Isolated system | II.     | Exchange of energy Only            |
| C.     | Open system     | III.    | No exchange of energy and matter   |
| D.     | Non-rigid wall  | IV.     | Immovable                          |
| E.     | Closed system   | V.      | Exchange of both energy and matter |

**A** A-IV, B-III, C-V, D-I, E-II

**B** A-I, B-III, C-IV, D-V, E-II

**C** A-I, B-II, C-IV, D-V, E-III

**D** A-II, B-III, C-IV, D-I, E-V

## Question

**An isolated system is that system in which;**

- A** there is no exchange of energy with the surroundings.
- B** there is exchange of mass and energy with the surroundings.
- C** there is no exchange of energy and mass with the surroundings.
- D** there is exchange of mass with the surroundings.

## Question

**If a closed system has adiabatic boundaries, then atleast one boundary must be;**

- A** permeable.
- B** imaginary.
- C** movable.
- D** fixed.

## Question

**A system that can exchange only energy but cannot exchange matter with the surrounding is known as;**

- A** open system.
- B** isolated system.
- C** closed system.
- D** None of these

## Question

**In thermodynamics, a quantity whose value simply depends upon the initial and final state of the system is called;**

- A** thermodynamic quantity.
- B** state function.
- C** adiabatic quantity.
- D** path function.

## Question

**A thermodynamic state function is;**

- A** one which obeys all the law of thermodynamics.
- B** a quantity which is used in measuring thermal changes.
- C** one which is used in thermochemistry.
- D** a quantity whose value depends only on the state of the system.

## Question

**Which of the following is not a state function?**

- A** Internal energy
- B** Free energy
- C** Work
- D** Enthalpy

## Question

**Which one of the following statements is false?**

- A** Work is a state function.
- B** Temperature is a state function.
- C** Change in the value of state function is completely defined when the initial and final states are specified.
- D** None of these.

## Question

**Enthalpy is a;**

- A** state function.
- B** path function.
- C** Both (A) and (B).
- D** None of these.

## Question

Which of the following sets contains only extensive properties?

- A** Mole, volume, pressure
- B**  $H$ ,  $U$ ,  $V$
- C**  $T$ ,  $P$ ,  $V$
- D** Density, entropy, heat capacity

## Question

**Which among the following state functions is an extensive property of the system**

- A** Temperature
- B** Volume
- C** Refractive index
- D** Viscosity

## Question

An ideal gas expands isothermally from  $10^{-3} \text{ m}^3$  to  $10^{-2} \text{ m}^3$  at 300 K against constant pressure of  $10^5 \text{ Nm}^{-2}$ . The work done on the gas is;

- A** +270 kJ
- B** -900 J
- C** +900 kJ
- D** -900 kJ

## Question

If  $x$  and  $y$  are arbitrary extensive variable, then

- A**  $(x + y)$  is an intensive variable
- B**  $\frac{x}{y}$  is intensive variable
- C**  $(x - y)$  is intensive variable
- D**  $(x \times y)$  is intensive variable

## Question

A gas expands from 3.0 L to 3.5 L against an external pressure of 1 atm. Calculate the PV work done.

**A** -0.5 L atm

**B** -1 L atm

**C** -100 J

**D** -200 J

## Question

**A thermally isolated gaseous system can exchange energy with the surroundings.  
The mode of transference of energy can be;**

- A** heat.
- B** work.
- C** heat and radiation.
- D** only radiations.

## Question

Quantity of work (in joules) done by the gas if it expands against a constant pressure of 0.980 atm and the change in volume ( $\Delta V$ ) is 25.0 L, is

- A** 24.5 J
- B** 2.48 J
- C**  $2.48 \times 10^3$  J
- D** 0.0245 J

## Question

**Changes in a system from an initial state to the final state were made by different manner that  $\Delta H$  remains same but  $q$  changes because:**

- A**  $\Delta H$  is a path function and  $q$  is a state function.
- B**  $\Delta H$  is a state function and  $q$  is a path function.
- C** Both  $\Delta H$  and  $q$  are state functions.
- D** Both  $\Delta H$  and  $q$  are path functions.

## Question

Which of the following is not a state function?

**A**  $\Delta S$

**B**  $\Delta G$

**C**  $\Delta H$

**D**  $q$

## Question

**Calculate the number of kJ of heat necessary to raise the temperature of 60.0 g of aluminium from 35°C to 55°C. Molar heat capacity of Al is  $24 \text{ J mol}^{-1} \text{ K}^{-1}$ .**

## Question

Calculate the enthalpy change on freezing of 1.0 mol of water at 10.0°C to ice at 10.0°C.  $\Delta_{\text{fus}}H = 6.03 \text{ kJ mol}^{-1}$  at 0°C.

$$C_p[\text{H}_2\text{O(l)}] = 75.3 \text{ J mol}^{-1} \text{ K}^{-1}$$

$$C_p[\text{H}_2\text{O(s)}] = 36.8 \text{ J mol}^{-1} \text{ K}^{-1}$$

## Question

Enthalpy of combustion of carbon to  $\text{CO}_2$  is  $-393.5 \text{ kJ mol}^{-1}$ . Calculate the heat released upon formation of 35.2 g of  $\text{CO}_2$  from carbon and dioxygen gas.

## Question

**Which of the following conditions must be satisfied for a system to be in thermal equilibrium?**

- A** The system must be isolated from its surroundings.
- B** The system must be at a constant temperature.
- C** The system must be at a constant pressure.
- D** The system must be at a constant energy state.

## Question

**A system is in thermodynamic equilibrium if it is in:**

- A** only thermal equilibrium.
- B** chemical, mechanical and thermal equilibrium.
- C** only chemical equilibrium.
- D** only mechanical equilibrium.

## Question

**Which of the following statements regarding latent heat is correct?**

- A** Latent heat depends on the mass of the substance.
- B** Latent heat is released during a phase change.
- C** Latent heat is measured in units of temperature.
- D** Latent heat is the heat required to change the temperature of a substance without changing its phase.

## Question

**Which of the following statements regarding heat is correct?**

- A** It is path independent quantity.
- B** It is path dependent quantity.
- C** Heat absorbed by the system is negative, i.e.,  $q < 0$ .
- D** Heat given out by the system is positive, i.e.,  $q > 0$ .

## Question

The work done when two moles of an ideal gas is compressed from a volume of  $5 \text{ m}^3$  to  $1 \text{ dm}^3$  at  $300 \text{ K}$ , under a pressure of  $100 \text{ kPa}$  is;

- A**  $499.9 \text{ kJ}$
- B**  $-499.9 \text{ kJ}$
- C**  $-99.5 \text{ kJ}$
- D**  $424.95 \text{ kJ}$

## Question

A system is changed from state A to state B by one path and from B to A by another path. If  $\Delta E_1$  and  $\Delta E_2$  are the corresponding changes in internal energy then:

- A**  $\Delta E_1 + \Delta E_2 = +ve$
- B**  $\Delta E_1 + \Delta E_2 = -ve$
- C**  $\Delta E_1 + \Delta E_2 = 0$
- D** None of these

## Question

**Which of the following statements is true regarding latent heat of fusion and latent heat of vaporization?**

- A** The latent heat of fusion is always greater than the latent heat of vaporization.
- B** Both latent heats are specific properties of a substance.
- C** Latent heat of fusion depends on pressure, while latent heat of vaporization does not.
- D** Latent heat of vaporization is the heat required to convert a gas into a liquid.

## Question

**A system absorb 600 J of heat and work equivalent to 300 J on its surroundings  
The change in internal energy is;**

- A** 300 J
- B** 400 J
- C** 500 J
- D** 600 J

## Question

Under isothermal condition, a gas at 300 K expands from 0.1 L to 0.25 L against a constant external pressure of 2 bar. The work done by the gas is; (Given that 1 L bar = 100 J)

**A** -30 J

**B** 5 kJ

**C** 25 J

**D** 30 J

## Question

If the internal energy of a system increases by 800 J and it releases 300 J of heat to its surroundings, how much work is done by the system?

- A** 500 J
- B** 1100 J
- C** 300 J
- D** -1100 J

## Question

The work done during the expansion of a gas from a volume of  $4 \text{ dm}^3$  to  $6 \text{ dm}^3$  against a constant external pressure of 3 atm, is;

- A**  $-6 \text{ J}$
- B**  $-608 \text{ J}$
- C**  $+304 \text{ J}$
- D**  $-304 \text{ J}$

## Question

A gas is allowed to expand in a well insulated container against a constant internal pressure of 2 atm from an initial volume of 2.50 L to a final volume of 4.50 L. The change in internal energy of the gas in joules will be:

- A** +404 J
- B** 1136.25 J
- C** -500 J
- D** -404 J

## Question

**A cup of tea placed in the room eventually acquires room temperature by losing heat. The process may be considered close to:**

- A** cyclic process.
- B** reversible process.
- C** isothermal process.
- D** None of these

## Question

**An irreversible process is characterized by;**

- A** the system returning to its initial state after completion of the process.
- B** a continuous decrease in temperature of the system.
- C** the inability to completely restore both the system and its surroundings to their initial states.
- D** the absence of any heat transfer during the process.

## Question

A gas expands isothermally against a constant external pressure of 1 atm from a volume of 10 dm<sup>3</sup> to a volume of 20 dm<sup>3</sup>. It absorbs 800 J of thermal energy from its surroundings. The  $\Delta U$  is;

- A** -312 J
- B** +123 J
- C** -213 J
- D** +231 J

## Question

**In which of the following processes, the work done by the system is equal to the heat added to the system?**

- A** Isothermal process
- B** Adiabatic process
- C** Isobaric process
- D** Isochoric process

## Question

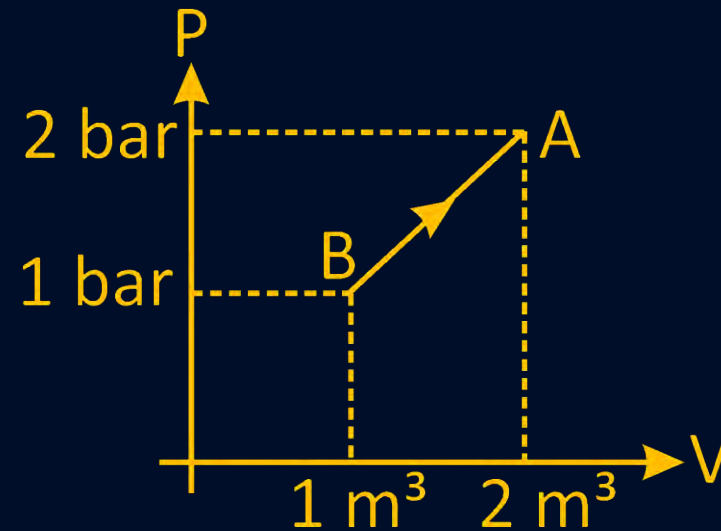
**Under what conditions can the First Law of Thermodynamics be applied?**

- A** Only in reversible processes.
- B** In any process, regardless of system boundaries.
- C** Only in closed systems.
- D** Only in adiabatic processes.

## Question

A system absorbs 100 kJ heat in the process shown in the figure. What is  $\Delta U$  for the system?

- A** -50 kJ
- B** +50 kJ
- C** +150 kJ
- D** -150 kJ



## Question

**Which of the following conditions is necessary for a process to be reversible?**

- A** The process must occur in an isolated system.
- B** The process must involve a change in temperature.
- C** The process must occur slowly and without friction.
- D** The process must lead to a decrease in temperature.

## Question

**During an isochoric process, 400 J of heat is added to a system. How much work is done by the system?**

- A** 0 J
- B** 400 J
- C** -400 J
- D** Cannot be determined

## Question

If 400 kJ work is done by the system and 150 kJ heat is given to system then, what will be the effect on internal energy?

- A** Increases by 250 kJ
- B** Decreases by 250 kJ
- C** Increases by 600 kJ
- D** Increases by 600 kJ

## Question

The first law of thermodynamics is represented by the equation;

**A**  $\Delta E = q + w$

**B**  $\Delta E = q - w$

**C**  $\Delta E = q$

**D**  $q = w - \Delta E$

## Question

When 1 mole gas is heated at constant volume, temperature is raised from 298 K to 308 K. Heat supplied to the gas is 800 J. Then, which statement is correct?

- A**  $q = w = 800 \text{ J}, \Delta E = 0$
- B**  $q = \Delta E = 800 \text{ J}, w = 0$
- C**  $q = -w = 800 \text{ J}, \Delta E = 0$
- D**  $\Delta E = 0, q = w = -800 \text{ J}$

## Question

Which of the following statements best describes the first law of thermodynamics?

- A** Energy cannot be created or destroyed, it can only change forms.
- B** Energy can be created or destroyed, but its total amount remains constant.
- C** Energy can be created or destroyed, but its total amount can change.
- D** Energy is conserved in isolated systems, but not in open systems.

## Question

For an isothermal expansion correct relationship between  $w_{\text{irr}}$  and  $w_{\text{rev}}$  (magnitude only):

- A**  $w_{\text{irr}} > w_{\text{rev}}$
- B**  $w_{\text{irr}} = w_{\text{rev}}$
- C**  $w_{\text{irr}} < w_{\text{rev}}$
- D** None of these

## Question

**In an isothermal expansion of an ideal gas:**

- A**  $q = 0$
- B**  $\Delta V = 0$
- C**  $\Delta U = 0$
- D**  $w = 0$

## Question

Which of the following is true for an adiabatic process?

**A**  $\Delta H = 0$

**B**  $w = 0$

**C**  $q = 0$

**D**  $\Delta V = 0$

## Question

Match List-I with List-II to find out the correct option.

| List-I |                                                         | List-II |                              |
|--------|---------------------------------------------------------|---------|------------------------------|
| A.     | A process carried out infinitesimally slowly.           | I.      | Adiabatic                    |
| B.     | A process in which no heat enters or leaves the system. | II.     | $\Delta E = 0, \Delta H = 0$ |
| C.     | A process carried out at constant temperature.          | III.    | Reversible                   |
| D.     | Cyclic process.                                         | IV.     | Isothermal                   |

**A** A-II, B-I, C-IV, D-III

**B** A-IV, B-II, C-I, D-III

**C** A-IV, B-I, C-III, D-II

**D** A-III, B-I, C-IV, D-II

## Question

**In which of the following processes, the pressure of an ideal gas remain constant**

- A** Isothermal expansion
- B** Isobaric compression
- C** Adiabatic expansion
- D** Isochoric heating

## Question

**Assertion (A):** An adiabatic process is a process in which the system does not exchange energy with its surroundings.

**Reason (R):** It does not involve any change in temperature of the system.

- A** Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A).
- B** Both Assertion (A) and Reason (R) are true but Reason (R) is not the correct explanation of Assertion (A).
- C** Assertion (A) is true and Reason (R) is false.
- D** Assertion (A) is false and Reason (R) is true.

## Question

**In which of the following processes, the work done by the system is equal to the heat added to the system?**

- A** Isothermal process
- B** Adiabatic process
- C** Isobaric process
- D** Isochoric process

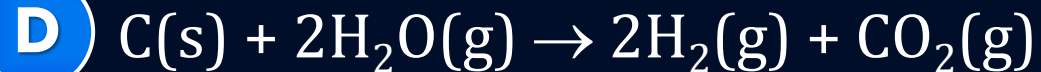
## Question

For which change  $\Delta H = \Delta U$ :



## Question

Assume each reaction is carried out in an open container. For which reaction will  $\Delta H = \Delta U$ ?



## Question

For the reaction,  $\text{C}_3\text{H}_5(\text{g}) + 5\text{O}_2(\text{g}) \rightarrow 3\text{CO}_2(\text{g}) + 4\text{H}_2\text{O}(\text{l})$  at constant temperature,  $\Delta H = \Delta U$  is

- A**  $+RT$
- B**  $-3RT$
- C**  $+3RT$
- D**  $-RT$

## Question

One mole of a non-ideal gas undergoes a change of state (2.0 atm, 3.0 L, 95 K)  $\rightarrow$  (4.0 atm, 5.0 L, 245 K) with a change in internal energy,  $\Delta U = 30.0 \text{ L atm}$ . The change in enthalpy ( $\Delta H$ ) is the process in L atm is:

- A** 40.0
- B** 42.3
- C** 44.0
- D** Not defined, because process is not constant

## Question

**The total number of intensive properties from the following is\_\_\_\_\_.**

**Volume, Molar heat capacity, Molarity,  $E^\theta$  cell, Gibbs free energy change, Molar mass, Mole**

## Question

Among the following the number of state variable is \_\_\_\_\_.

Internal energy (U)

Heat (q)

Volume (V)

Enthalpy (H)

## Question

What happens when methane undergoes combustion in systems A and B respectively?

**Adiabatic system  
System A**

- A** Temperature rises
- B** Temperature falls
- C** Temperature falls
- D** Temperature remains same

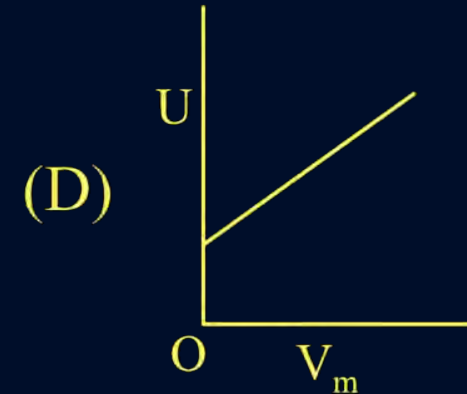
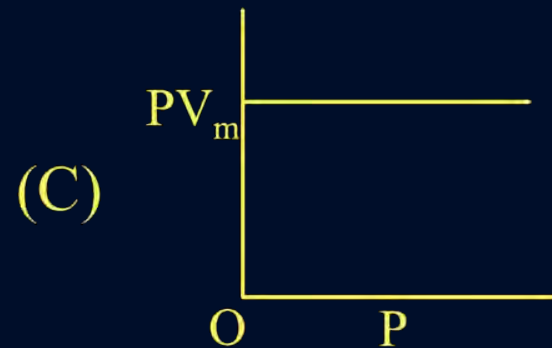
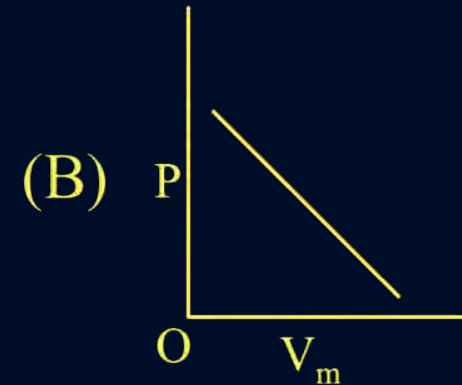
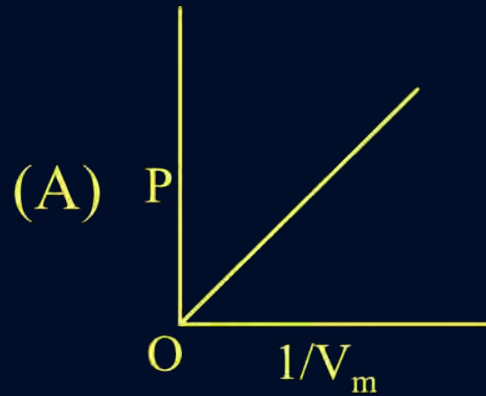
**Diathermic container  
System B**

- Temperature remains same
- Temperature rises
- Temperature remains same
- Temperature rises

## Question

The combination of plots which do not represent isothermal expansion of an ideal gas is:

- A** (B) and (D)
- B** (A) and (C)
- C** (B) and (C)
- D** (A) and (D)



## Question

One mole of an ideal gas expands isothermally and reversibly from  $10 \text{ dm}^3$  to  $20 \text{ dm}^3$  at  $300 \text{ K}$ .  $\Delta U$ ,  $q$  and work done in the process respectively are:

Given:  $R = 8.3 \text{ JK}^{-1} \text{ mol}^{-1}$

$$\ln 10 = 2.3$$

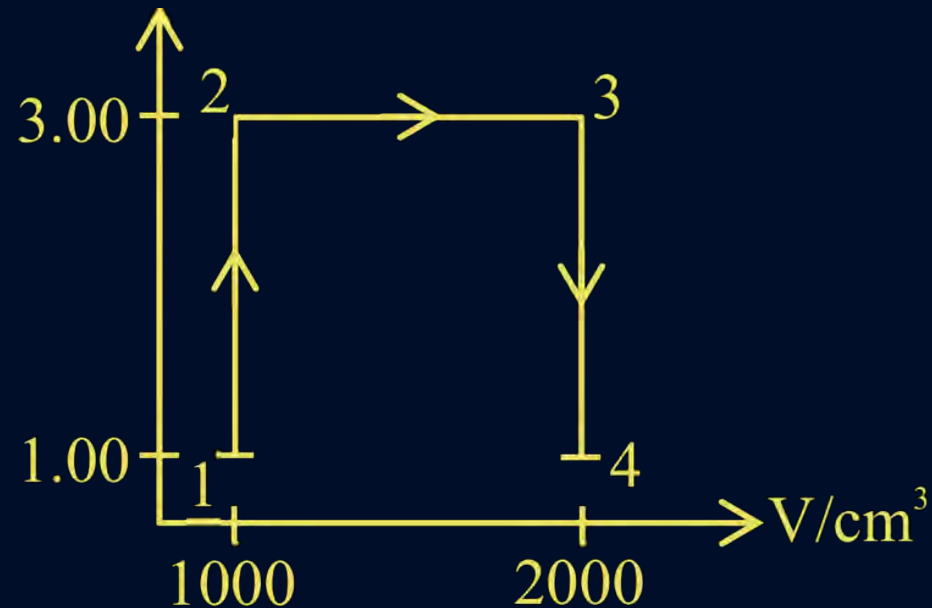
$$\log 2 = 0.30$$

$$\log 3 = 0.48$$

- A**  $0, 21.84 \text{ kJ}, -1.726 \text{ kJ}$
- B**  $0, -17.18 \text{ kJ}, 1.718 \text{ J}$
- C**  $0, 21.84 \text{ kJ}, 21.84 \text{ kJ}$
- D**  $0, 1.718 \text{ kJ}, -1.718 \text{ kJ}$

## Question

**P/atm**



A perfect gas (0.1 mol) having  $\bar{C}_v = 1.50 R$  (independent of temperature) undergoes the above transformation from point 1 to point 4. If each step is reversible, the total work done (w) while going from point 1 to point 4 is (-) \_\_\_ J (nearest integer)

[Given :  $R = 0.082 \text{ L atm K}^{-1} \text{ mol}^{-1}$ ]

## Question

Arrange the following in order of magnitude of work done by the system/on the system at constant temperature:

- A.  $|W_{\text{reversible}}|$  for expansion in infinite stage.
- B.  $|W_{\text{irreversible}}|$  for expansion in single stage.
- C.  $|W_{\text{reversible}}|$  for compression in infinite stage.
- D.  $|W_{\text{irreversible}}|$  for compression in single stage.

Choose the correct answer from the options given below:

**A**  $A > B > C > D$

**B**  $D > C = A > B$

**C**  $C = A > D > B$

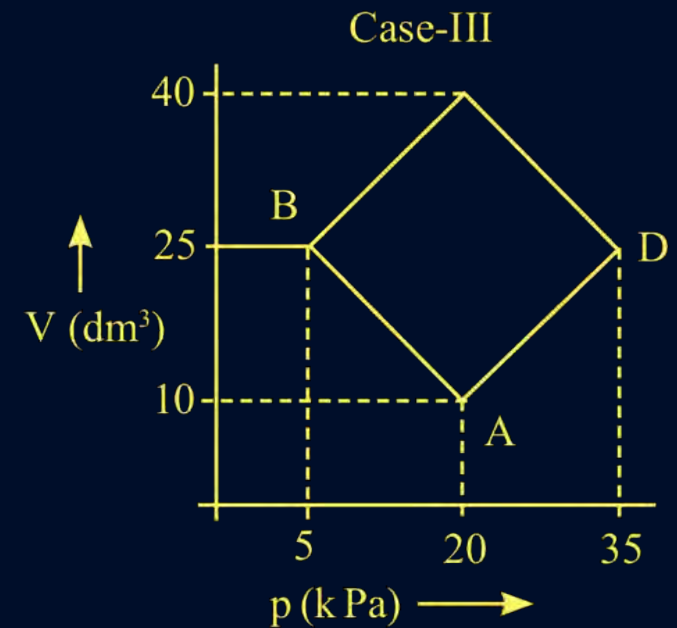
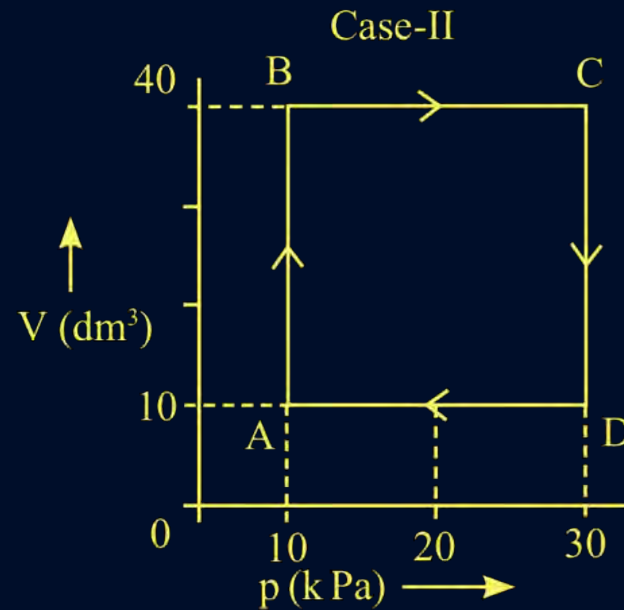
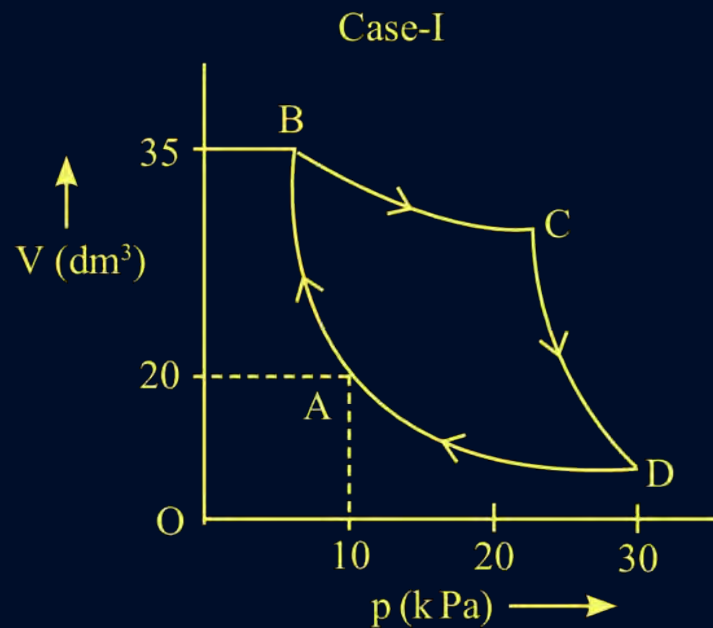
**D**  $A > C > B > D$

## Question

A liquid when kept inside a thermally insulated closed vessel at 25°C was mechanically stirred from outside. What will be the correct option for the following thermodynamic parameters?

- A**  $\Delta U = 0, q < 0, w > 0$
- B**  $\Delta U > 0, q = 0, w > 0$
- C**  $\Delta U < 0, q = 0, w > 0$
- D**  $\Delta U = 0, q = 0, w = 0$

## Question



An ideal gas undergoes a cyclic transformation starting from the point A and coming back to the same point by tracing the path  $A \rightarrow B \rightarrow C \rightarrow D \rightarrow A$  as shown in the three cases above. Choose the correct option regarding  $\Delta U$ :

**A**  $\Delta U \text{ (Case-III)} > \Delta U \text{ (Case-II)} > \Delta U \text{ (Case-I)}$

**B**  $\Delta U \text{ (Case-I)} = \Delta U \text{ (Case-II)} = \Delta U \text{ (Case-III)}$

**C**  $\Delta U \text{ (Case-I)} > \Delta U \text{ (Case-II)} > \Delta U \text{ (Case-III)}$

**D**  $\Delta U \text{ (Case-I)} > \Delta U \text{ (Case-III)} > \Delta U \text{ (Case-II)}$

## Question

**500 J of energy is transferred as heat to 0.5 mol of Argon gas at 298 K and 1.00 atm. The final temperature and the change in internal energy respectively are:  
Given:  $R = 8.3 \text{ J K}^{-1} \text{ mol}^{-1}$**

- A** 378 K and 300 J
- B** 348 K and 300 J
- C** 368 K and 500 J
- D** 378 K and 500 J

## Question

If three moles of an ideal gas at 300 K expand isothermally from 30 dm<sup>3</sup> to 45 dm<sup>3</sup> against a constant opposing pressure of 80 kPa, then the amount of heat transferred is\_\_\_\_\_ J.

## Question

Which amongst the following options is the correct relation between change in enthalpy and change in internal energy? [2023]

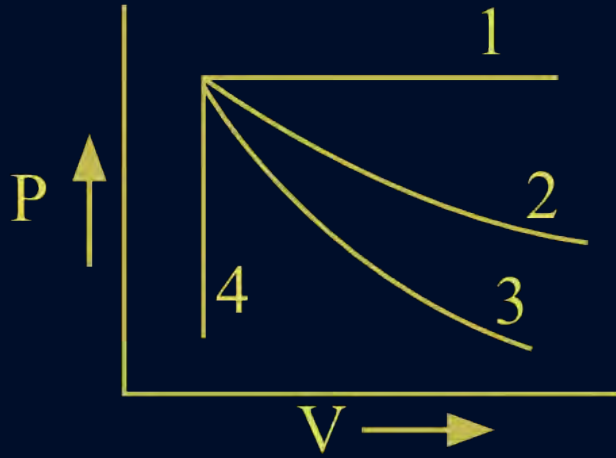
**A**  $\Delta H + \Delta U = \Delta nR$

**B**  $\Delta H = \Delta U - \Delta n_g RT$

**C**  $\Delta H = \Delta U + \Delta n_g RT$

**D**  $\Delta H - \Delta U = -\Delta nRT$

## Question



**Isobaric process is represented by:**

**A** 3

**B** 4

**C** 1

**D** 2

## Question

**Match List-I with List-II to find out the correct option.**

**A** A-III, B-I, C-IV, D-II

**B** A-IV, B-II, C-I, D-III

**C** A-IV, B-I, C-III, D-II

**D** A-I, B-III, C-IV, D-II

| List-I |            | List-II |                       |
|--------|------------|---------|-----------------------|
| (A)    | Adiabatic  | (I)     | $\Delta V = 0$        |
| (B)    | Isochoric  | (II)    | $\Delta T = 0$        |
| (C)    | Isobaric   | (III)   | $q = 0$               |
| (D)    | Isothermal | (IV)    | $P = \text{constant}$ |

## Question

A gas expands adiabatically at constant pressure such that  $TV^{1/2}$ . The value of  $(C_p/C_v)$  of the gas will be:

- A** 1.30
- B** 1.50
- C** 1.70
- D** 2

## Question

One mole of an ideal gas at 300 K is expanded isothermally from 1 L to 10 L volume.  $\Delta U$  for this process is: (2022)

- A** 1260 J
- B** 2520 J
- C** 5040 J
- D** 0 J

## Question

The maximum work obtained by an isothermal reversible expansion of 1 mole of an ideal gas at  $27^{\circ}\text{C}$  from 2.24 to 22.4 L is ( $R = 2 \text{ cal K}^{-1} \text{ mol}^{-1}$ )

**A** -1381.8 cal

**B** -600 cal

**C** -138.18 cal

**D** -690.9 cal

## Question

For a gaseous reaction;  $\text{N}_2\text{O}_4(\text{g}) \rightarrow 2\text{NO}_2(\text{g})$

**A**  $\Delta H$  and  $\Delta U$  are equal.

**B**  $\Delta H - \Delta U = RT$

**C**  $\Delta H - \Delta U = 2R$

**D**  $\Delta H - \Delta U = 0$

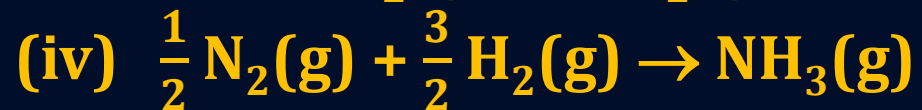
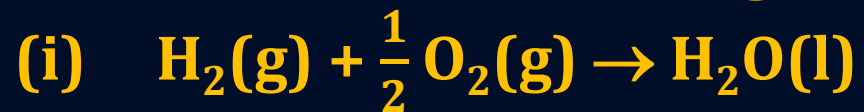
## Question

The latent heat of vapourisation of a liquid at 500 K and 1 atm pressure is 10. kcal/mol. What will be the change in internal energy of 3 moles of the liquid at the same temperature and pressure?

- A** 27.0 kcal
- B** 13.0 kcal
- C** -27.0 kcal
- D** -13.0 kcal

## Question

For which of the following reactions,  $\Delta H$  and  $\Delta U$  are equal?



**A** (i) only

**B** (ii) and (iii) only

**C** (iii) only

**D** (iv) only

## Question

The maximum work obtained by an isothermal reversible expansion of 1 mole of an ideal gas at  $27^{\circ}\text{C}$  from 2.24 to 22.4 L is ( $R = 2 \text{ cal K}^{-1} \text{ mol}^{-1}$ )

- A** -1381.8 cal
- B** -600 cal
- C** -138.18 cal
- D** -690.9 cal

## Question

2 mole of an ideal gas at  $27^{\circ}\text{C}$  expands isothermally and reversibly from volume of 4 litres to 40 litres. The work done (in kJ) is:

**A**  $w = -28.72 \text{ kJ}$

**B**  $w = -11.488 \text{ KJ}$

**C**  $w = -5.736 \text{ kJ}$

**D**  $w = -4.988 \text{ kJ}$

## Question

Two moles of an ideal monoatomic gases are allowed to expand adiabatically and reversibly from 300 K to 200 K. The work done in the system is ( $C_v = 12.5 \text{ J/K mol}$ )

- A** -12.5 kJ
- B** -2.5 kJ
- C** -625 kJ
- D** 500 kJ

## Question

What is the final temperature of 0.10 mole monoatomic ideal gas that performs 75 cal of work adiabatically if the initial temperature is  $227^{\circ}\text{C}$ ? (use  $R = 2 \text{ cal/K mol}$ )

**A** 250 K

**B** 300 K

**C** 350 K

**D** 750 K

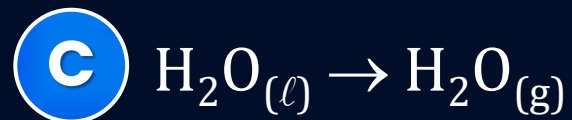
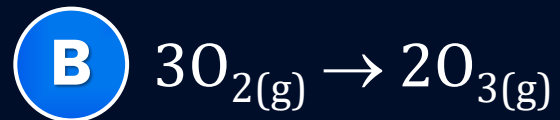
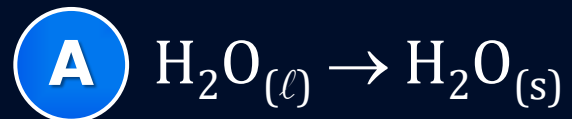
## Question

**Which of the following processes are accompanied by increase of entropy:**

- (i) Dissolution of iodine in a solvent  $\{I_2(s) \rightarrow I_2(aq.)\}$**
- (ii) HCl is added to  $AgNO_3$  and a precipitate of AgCl is obtained.**
- (iii) A partition is removed to allow two gases to mix.**

## Question

In which reaction  $\Delta S$  is positive:



## Question

Which of the following reactions is associated with negative change in entropy?



## Question

If  $S^\circ$  for  $\text{H}_2$ ,  $\text{Cl}_2$  and  $\text{HCl}$  are  $0.13$ ,  $0.22$  and  $0.19 \text{ KJ K}^{-1} \text{ mol}^{-1}$  respectively. The total change in standard entropy for the reaction  $\text{H}_2 + \text{Cl}_2 \longrightarrow 2\text{HCl}$  is:

- A**  $30 \text{ JK}^{-1} \text{ mol}^{-1}$
- B**  $40 \text{ JK}^{-1} \text{ mol}^{-1}$
- C**  $60 \text{ JK}^{-1} \text{ mol}^{-1}$
- D**  $20 \text{ JK}^{-1} \text{ mol}^{-1}$

## Question

The enthalpy change for transition of liquid water to steam is  $40.8 \text{ kJ mol}^{-1}$  at  $373\text{K}$ . Calculate  $\Delta S$  for the process.

## Question

**Calculate the change in entropy for the fusion of 1 mol of ice. The melting point of ice is 273K and molar enthalpy of fusion of ice =  $6 \text{ kJ mol}^{-1}$ .**

## Question

The enthalpy of vapourisation of liquid diethyl ether  $(\text{C}_2\text{H}_5)_2\text{O}$ , is  $26.0 \text{ kJ mol}^{-1}$  at its boiling point ( $35.0^\circ\text{C}$ ). Calculate  $\Delta S$  for conversion of:

- (i) Liquid to vapour and
- (ii) Vapour to liquid at  $35^\circ\text{C}$

## Question

Among the following the set of parameters that represents path function is:

(A)  $q + w$

(B)  $w$

(C)  $q$

(D)  $H - TS$

**A** (B) and (C)

**B** (B), (C) and (D)

**C** (A), (B) and (C)

**D** (A) and (D)

## Question

A 5 litre cylinder contained 10 mol of oxygen gas at 27°C. Due to sudden leakage through the hole, all the gas escaped into the atmosphere and the cylinder got empty. If the atmosphere pressure is 1.0 atm. Find work (in L atm) ( $R = 0.083 \text{ L atm mol}^{-1} \text{ K}^{-1}$ )

## Question

**Find the work in each case:**

- (a) When one mol of ideal gas in 10 litre container at 1 atm is allowed to enter a vacuum bulb of capacity 100 litre.**
- (b) When 1 mol of gas expands from 1 litre to 5 litre against constant atmospheric pressure.**

## Question

1g of water changes from liquid to vapour phase at constant pressure of 1 atmosphere, the volume increases from 1 mL to 1671 mL. The heat of vaporisation at this pressure is  $540 \text{ Cal g}^{-1}$ . Find the increase in internal energy of water. ( $1 \text{ L atm} = 101 \text{ J}$ )

## Question

A gas occupies 2 L at STP. It is provided 300 J heat so that its volume become 2.5 L at 1 atm. Calculate change in its internal energy.

## Question

A sample of gas present in a cylinder fitted with a frictionless piston expands against a constant pressure of 1 atm from a volume of 2L to 12L. During the process, it absorbs 600 J of heat from the surroundings. Calculate the change in internal energy of the system.

## Question

Two moles of an ideal gas at 2 atm and 27°C is compressed isothermally to one half of its volume by a constant external pressure of 4 atm. Calculate  $q$ ,  $w$  and  $\Delta E$  ( $R = 0.082 \text{ L atm mol}^{-1} \text{ K}^{-1}$ )

## Question

The enthalpy of vaporization for water is  $186.5 \text{ KJ mol}^{-1}$ , the entropy of its vaporization will be-

- A**  $0.5 \text{ KJ K}^{-1} \text{ mol}^{-1}$
- B**  $1.0 \text{ KJ K}^{-1} \text{ mol}^{-1}$
- C**  $1.5 \text{ KJ K}^{-1} \text{ mol}^{-1}$
- D**  $2.0 \text{ KJ K}^{-1} \text{ mol}^{-1}$

## Question

**Calculate the entropy change in melting of one gm ice at  $0^{\circ}\text{C}$  if latent heat of ice is  $80 \text{ cal/g}$ -**

**A**  $80 \text{ Cal K}^{-1}$

**B**  $20 \text{ Cal K}^{-1}$

**C**  $4.4 \text{ Cal K}^{-1}$

**D**  $0.3 \text{ Cal K}^{-1}$

## Question

5 mole of an ideal gas expand reversibly from a volume of  $8\text{ d m}^3$  to  $80\text{ d m}^3$  at a temperature of  $27^\circ\text{C}$ . The change in entropy is:

- A**  $41.57 \text{ JK}^{-1}$
- B**  $-95.73 \text{ JK}^{-1}$
- C**  $95.73 \text{ JK}^{-1}$
- D**  $-41.57 \text{ JK}^{-1}$

## Question

1 mole of a diatomic ideal gas at  $25^{\circ}\text{C}$  is subjected to expand reversibly and adiabatically to ten times of its initial volume. Calculate the change in entropy during expansion (in  $\text{JK}^{-1} \text{mol}^{-1}$ )

- A**  $R \ln 10$
- B**  $-R \ln 10$
- C**  $2.5 R \ln 10$
- D** Zero

## Question

If one mole of an ideal gas is expanded isothermally at 300 K until it's volume is tripled, then change in entropy of gas is:

- A** Zero
- B** Infinity
- C**  $\frac{5}{2}R \ln 3$
- D**  $R \ln 3$

## Question

**In a spontaneous irreversible process the total entropy of the system and surroundings**

- A** Remains constant
- B** Increases
- C** Decreases
- D** Zero

## Question

**Match the column:**

|    | Column – I           |      | Column – II    |
|----|----------------------|------|----------------|
| a. | Adiabatic process    | i.   | $q = 0$        |
| b. | Isothermal process   | ii.  | $\Delta H = 0$ |
| c. | Isoenthalpic process | iii. | $\Delta T = 0$ |
| d. | Isoentropic process  | iv.  | $\Delta S = 0$ |

**A** a-(i); b-(iv); c-(ii); d-(iii)

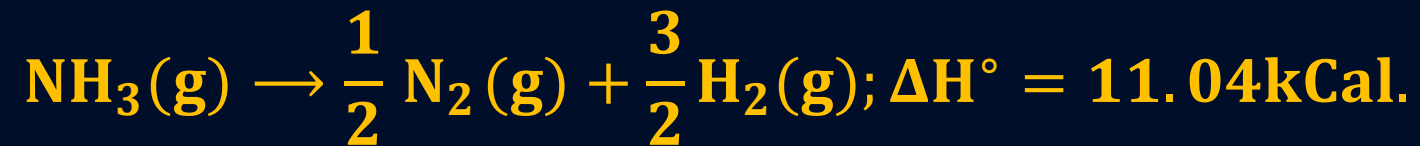
**B** a-(ii); b-(i); c-(iv); d-(iii)

**C** a-(i); b-(iii); c-(ii); d-(iv)

**D** a-(i); b-(iii); c-(iv); d-(ii)

## Question

For the reaction at 25°C



Calculate  $\Delta E^\circ$  of the reaction at the given temperature.

## Question

At 27°C the internal energy change of reaction  $\text{H}_2(\text{g}) + \text{Cl}_2(\text{g}) \rightarrow 2\text{HCl}(\text{g})$  is 2Cal. What is the enthalpy change of this reaction.

## Question

The heat of combustion of gaseous methane ( $\text{CH}_4$ ) at constant volume is measured in bomb calorimeter at 298K is found to be  $-885.4 \text{ kJ mol}^{-1}$ . Find the value of enthalpy change at the same temperature.

## Question

5 moles of oxygen are heated at constant volume from  $10^{\circ}\text{C}$  to  $20^{\circ}\text{C}$ . What will be the change in the internal energy of gas? The molar heat capacity of oxygen at constant pressure,  $C_p = 7.03 \frac{\text{Cal}}{\text{molK}}$  and  $R = 2 \text{ Cal mol}^{-1} \text{ K}^{-1}$

## Question

At 27°C, one mole of an ideal gas compressed isothermally and reversibly from a pressure of 2 atm to 10 atm. Calculate  $\Delta E$  and  $q$  in calorie.

## Question

A gas expands from  $3 \text{ dm}^3$  to  $5 \text{ dm}^3$  against a constant pressure of  $3 \text{ atm}$ . The work done during expansion is used to heat  $10 \text{ mol}$  of water of temperature  $290 \text{ K}$ . Calculate final temperature of water (if specific heat of water is  $4.184 \text{ J g}^{-1}\text{K}^{-1}$ )

## Question

A sample of 3 mol of an ideal gas at 200K and 2 atm is compressed reversibly and adiabatically until the temperature reaches 250K, given that molar heat capacity is  $27.5 \text{ J K}^{-1} \text{ mol}^{-1}$  at constant volume, calculate w.

## Question

Which of the following are state function?

- (i)  $q$
- (ii) Entropy
- (iii) Gibb's energy
- (iv)  $H$
- (v)  $w$

## Question

10 moles of an ideal gas at  $27^{\circ}\text{C}$  and 10 atm. pressure occupying a volume of 24.6 L undergoes the following changes.

- (i) Isothermal and reversible expansion to 246 L
- (ii) Isothermal and irreversible expansion to 246 L.
- (iii) Isochoric heating to  $177^{\circ}\text{C}$ .

Calculate the work in each transformation in kJ.

## Question

Find the work, when 2 mol of a gas expands isothermally from  $5\text{dm}^3$  to  $40\text{dm}^3$  against a constant external pressure of 2 atm at 298K.  
Also calculate  $w_{\text{rev}}$  for the change.

## Question

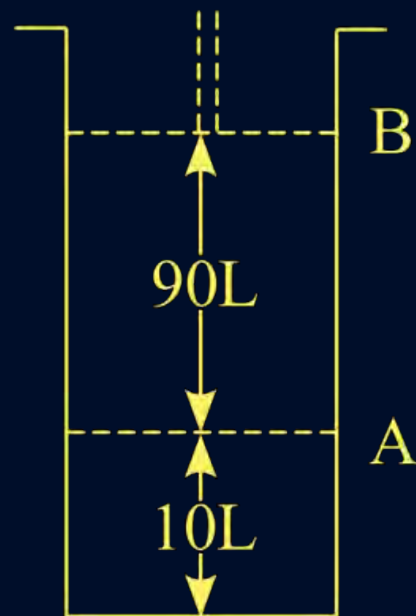
One mole of an ideal gas expands isothermally and reversibly from  $10 \text{ dm}^3$  to  $20 \text{ dm}^3$  at  $300 \text{ K}$ .  $U$ ,  $q$  and work done in the process respectively are:

Given:  $R = 8.3 \text{ JK}^{-1} \text{ mol}^{-1}$

$\ln 10 = 2.3$   $\log 2 = 0.30$   $\log 3 = 0.48$

- A**  $0, 21.84 \text{ kJ}, -1.726 \text{ kJ}$
- B**  $0, -17.18 \text{ kJ}, 1.718 \text{ J}$
- C**  $0, 21.84 \text{ kJ}, 21.84 \text{ kJ}$
- D**  $0, 1.718 \text{ kJ}, -1.718 \text{ kJ}$

## Question



Consider the figure provided.

1 mol of an ideal gas is kept in a cylinder, fitted with a piston, at the position A at  $18^{\circ}\text{C}$ . If the piston is moved to position B, keeping the temperature unchanged, then 'x' L atm work is done in this reversible process.

x = \_\_\_\_\_ L atm. (nearest integer)

[Given: Absolute temperature =  $^{\circ}\text{C} + 273.15$ ,  $R = 0.08206 \text{ L atm mol}^{-1} \text{ K}^{-1}$ ]

## Question

For a reaction at 25°C enthalpy change ( $\Delta H$ ) and entropy change ( $\Delta S$ ) are  $-11.7 \times 10^3 \text{ J mol}^{-1}$  and  $-105 \text{ J mol}^{-1} \text{ K}^{-1}$  respectively. Find out whether this reaction is spontaneous or not.

## Question

Calculate the equilibrium constant for the reaction given below at 400K. If

$$\Delta H^\circ = 77.2 \text{ kJ mol}^{-1} \text{ and } \Delta S^\circ = 122 \text{ JK}^{-1} \text{ mol}^{-1}$$



## Question

For the reaction at 300 K.



$$\Delta U = -3.0 \text{ kcal} \quad \Delta S = -10.0 \text{ cal/K}$$

$$(R \approx 2 \text{ cal mol}^{-1} \text{ K}^{-1})$$

$\Delta G$  is:

- ☒ A -600 cal
- ☐ B -3600 cal
- ☐ C 2400 cal
- ☐ D 3000 cal

## Question

**For a spontaneous process:**

- A**  $\Delta G = 0$
- B**  $\Delta G < 0$
- C**  $\Delta G > 0$
- D** Any of the above

## Question

The value of  $\Delta H$  and  $\Delta S$  for the reaction

$\text{C}_{\text{graphite}} + \text{O}_2(\text{g}) \rightarrow \text{CO}_2(\text{g})$  are  $-100 \text{ kJ}$  and  $-100 \text{ JK}^{-1}$  respectively.

The reaction will be spontaneous at:

- A** 1000 K
- B** 900 K
- C** 1100 K
- D** At any temperature

## Question

Standard entropy of  $\text{N}_2$ ,  $\text{H}_2$  and  $\text{NH}_3$  is are 60, 40 and 50  $\text{JK}^{-1} \text{mol}^{-1}$  respectively. For the reaction  $\frac{1}{2}\text{N}_2 + \frac{3}{2}\text{H}_2 \rightarrow \text{NH}_3$ ,  $\Delta H = -30 \text{ kJ}$  to be at equilibrium, the temperature should be:

**A** 500 K

**B** 750 K

**C** 1000 K

**D** 1250 K

## Question

In which of the following cases, the reaction is spontaneous at all temperatures?

**A**  $\Delta H > 0, \Delta S > 0$

**B**  $\Delta H < 0, \Delta S > 0$

**C**  $\Delta H < 0, \Delta S < 0$

**D**  $\Delta H > 0, \Delta S < 0$

## Question

**A reaction occur spontaneously. If**

- A**  $T\Delta S = \Delta H$  and both  $\Delta H$  and  $\Delta S$  are positive
- B**  $T\Delta S > \Delta H$  and both  $\Delta H$  and  $\Delta S$  are positive
- C**  $T\Delta S > \Delta H$  and both  $\Delta H$  and  $\Delta S$  are negative
- D**  $T\Delta S = \Delta H$  and both  $\Delta H$  is positive and  $\Delta S$  is negative

## Question

**The equilibrium constant for a reaction is 10. What will be the value of  $\Delta G$ ?**

**$R = 8.314 \text{ J K}^{-1} \text{ mol}^{-1}$ ,  $T = 300 \text{ K}$**

## Question

For the water gas reaction



The standard Gibb's energy of reaction (at 1000 K) is  $-8.1 \text{ kJ mol}^{-1}$ . Value of equilibrium constant is–

**A** 2.6

**B** 6.2

**C** 8.2

**D** 10

## Question

Calculate  $\Delta G^\circ$  for the conversion of oxygen to ozone,



If  $K_p$  for this conversion is  $3 \times 10^{-29}$ .

- A**  $+175.3 \text{ kJ mol}^{-1}$
- B**  $+162.7 \text{ kJ mol}^{-1}$
- C**  $-162.7 \text{ kJ mol}^{-1}$
- D**  $-140.5 \text{ kJ mol}^{-1}$

## Question

A hypothetical reaction,  $A \rightarrow B$ , proceed through following sequence of steps-

A        C;         $\Delta H = q_1$

C        D;         $\Delta H = q_2$

D        B;         $\Delta H = q_3$

The heat of reaction s:

**A**  $q_1 - q_2 + 2q_3$

**B**  $q_1 + q_2 - q_3$

**C**  $q_1 + q_2 + q_3$

**D**  $q_1 + 2q_2 - 2q_3$

## Question

The heat of reaction for  $A + \frac{1}{2}O_2 \rightarrow AO$  is  $-50$  kcal/mol and  $AO + \frac{1}{2}O_2 \rightarrow AO_2$  is  $100$  kcal/mol. The heat of reaction (in kcal/mol) for  $A + O_2 \rightarrow AO_2$  will be:

**A**  $-50$

**B**  $+50$

**C**  $100$

**D**  $150$

## Question

Consider the following process:



For  $B + D \rightarrow G + 2C$ ;  $\Delta H$  will be

**A** +525 kJ

**B** +325 kJ

**C** -175 kJ

**D** -325 kJ

## Question

The heat of formation of  $\text{Fe}_2\text{O}_3(\text{s})$  is  $-824.2 \text{ kJ mol}^{-1}$ .

$\Delta H$  for the reaction.

$2\text{Fe}_2\text{O}_3(\text{s}) \rightarrow 4\text{Fe}(\text{s}) + 3\text{O}_2(\text{g})$  is:

**A**  $-412.1 \text{ kJ}$

**B**  $-1648.4 \text{ kJ}$

**C**  $-3296.8 \text{ kJ}$

**D**  $1648.4 \text{ kJ}$

## Question

The  $\Delta H^\circ$  for the reaction,

$4\text{S}(\text{s}) + 6\text{O}_2(\text{g}) \rightarrow 4\text{SO}_3(\text{g})$  is  $-1583.2 \text{ kJ}$ .

Standard enthalpy of formation of sulphur trioxide is:

- A**  $-3166.4 \text{ kJ}$
- B**  $3166.4 \text{ kJ}$
- C**  $-395.8 \text{ kJ}$
- D**  $395.8 \text{ kJ}$

## Question

$\Delta H$  for the reaction,



If the enthalpy of formation of  $\text{SO}_3(\text{g})$  is  $-395.4 \text{ kJ}$  then the enthalpy of formation of  $\text{SO}_2(\text{g})$  is:

**A**  $-297.1 \text{ kJ}$

**B**  $493.7 \text{ kJ}$

**C**  $-493.7 \text{ kJ}$

**D**  $297.1 \text{ kJ}$

## Question

**The enthalpies of all elements in their standard states are:**

- (i) unity**
- (ii) zero**
- (iii)  $< 0$**
- (iv) different for each element**

## Question

Which of the reaction defines molar  $\Delta H_f^\circ$ ?

- A**  $\text{CaO(s)} + \text{CO}_2\text{(g)} \rightarrow \text{CaCO}_3\text{(s)}$
- B**  $\frac{1}{2}\text{Br}_2\text{(g)} + \frac{1}{2}\text{H}_2\text{(g)} \rightarrow \text{HBr(g)}$
- C**  $\text{N}_2\text{(g)} + 2\text{H}_2\text{(g)} + \frac{3}{2}\text{O}_2\text{(g)} \rightarrow \text{NH}_4\text{NO}_3\text{(s)}$
- D**  $\text{I}_2\text{(g)} + \text{H}_2\text{(g)} \rightarrow 2\text{HI(g)}$

## Question

Calculate the heat of formation of  $\text{PCl}_5(\text{s})$  from the following data:



**A** -108.7 kcal

**B** 108.7 kcal

**C** -184.6 kcal

**D** 184.6 kcal

## Question

The standard enthalpies of formation of CO, NO<sub>2</sub> and CO<sub>2</sub> are -110.5 kJ mol<sup>-1</sup>, -33.2 kJ mol<sup>-1</sup> and -393.5 kJ mol<sup>-1</sup> respectively. The standard enthalpy of the reaction is:



**A** -1065.6 kJ

**B** -200 kJ

**C** -700 kJ

**D** 850 kJ

## Question

The enthalpies of combustion of carbon and carbon monoxide are  $-393.5$  and  $-283 \text{ kJ mol}^{-1}$  respectively. The enthalpy of formation of carbon monoxide per mole-

- A**  $110.5 \text{ kJ}$
- B**  $676.5 \text{ kJ}$
- C**  $-676.5 \text{ kJ}$
- D**  $-110.5 \text{ kJ}$

## Question

The enthalpy of combustion of methane, graphite and dihydrogen at 298 K are  $-890.3 \text{ kJ mol}^{-1}$ ,  $-393.5 \text{ kJ mol}^{-1}$ , and  $-285.8 \text{ kJ mol}^{-1}$  respectively. Enthalpy of formation of  $\text{CH}_4(\text{g})$  will be:

- A**  $-74.8 \text{ kJ mol}^{-1}$
- B**  $-52.27 \text{ kJ mol}^{-1}$
- C**  $+74.8 \text{ kJ mol}^{-1}$
- D**  $+52.26 \text{ kJ mol}^{-1}$

## Question

The enthalpy of formation for  $\text{C}_2\text{H}_4(\text{g})$ ,  $\text{CO}_2(\text{g})$  and  $\text{H}_2\text{O}(\text{l})$  at  $25^\circ\text{C}$  and 1 atm pressure be 52,  $-394$  and  $-286 \text{ kJ mol}^{-1}$  respectively. The enthalpy of combustion of  $\text{C}_2\text{H}_4(\text{g})$  will be-

- A**  $+1412 \text{ kJ mol}^{-1}$
- B**  $-1412 \text{ kJ mol}^{-1}$
- C**  $+141.2 \text{ kJ mol}^{-1}$
- D**  $-141.2 \text{ kJ mol}^{-1}$

## Question

The heat of combustion of solid benzoic acid at constant volume is  $-321.30 \text{ kJ}$  at  $27^\circ\text{C}$ . The heat of combustion at constant pressure is-

- A**  $-321.30 - 300 R$
- B**  $-321.30 + 300 R$
- C**  $-321.30 - 150 R$
- D**  $-321.30 + 900 R$

## Question

The enthalpy of vaporization of liquid water using data in  $\text{kJ mol}^{-1}$  is

Given:



**A** -527.61

**B** -43.93

**C** +527.61

**D** 43.93

## Question

The heat of combustion of yellow phosphorus and red phosphorus are  $-9.91 \text{ KJ/mol}$  and  $-8.78 \text{ KJ/mol}$  respectively. The heat of transition from yellow phosphorus to red phosphorus is-

- A**  $-1.13 \text{ KJ}$
- B**  $-18.69 \text{ KJ}$
- C**  $+18.69 \text{ KJ}$
- D**  $+1.13 \text{ KJ}$

## Question

Calculate the standard enthalpy of formation of  $\text{CH}_3\text{OH}(\text{l})$  from the following data:



## Question

The molar heat of formation of  $\text{NH}_4\text{NO}_3$  is  $-367.54 \text{ kJ}$  and those of  $\text{N}_2\text{O}(\text{g})$  and  $\text{H}_2\text{O}(\text{l})$  are  $81.46 \text{ kJ}$  and  $-285.78 \text{ kJ}$  respectively at  $25^\circ\text{C}$  and at 1 atmospheric pressure. Calculate  $H$  for the reaction



**A**  $-122.56 \text{ kJ}$

**B**  $-66.56 \text{ kJ}$

**C**  $112.56 \text{ kJ}$

**D**  $102.56 \text{ kJ}$

## Question



The enthalpy of formation of  $\text{H}_2\text{SO}_4$  at 298 K will be

**A**  $-433.7 \text{ kJ}$

**B**  $-650.3 \text{ kJ}$

**C**  $+320.5 \text{ kJ}$

**D**  $-813.9 \text{ kJ}$

## Question

From the given information, what is the standard enthalpy of formation for  $\text{Al}_2\text{O}_3(\text{s})$ ?



**A**  $-6704 \text{ kJ/mol}$

**B**  $-3352 \text{ kJ/mol}$

**C**  $-1676 \text{ kJ/mol}$

**D**  $1676 \text{ kJ/mol}$

## Question

The following reactions are given:



Calculate the heat of formation of  $\text{CH}_4$ ?

**A**  $x + y + z$

**B**  $y + 2z - x$

**C**  $x - y - 2z$

**D** None of the above

## Question

The Bond-energies of C – C, C–H, H–H and C = C are 198, 98, 103, 145 kcal respectively. The enthalpy change of the reaction

$\text{HC}=\text{CH} + \text{H}_2 \rightarrow \text{C}_2\text{H}_4$  is-

**A** –152 kcal

**B** 96 kcal

**C** 48 kcal

**D** –40 kcal

## Question

Heat evolved in the reaction  $\text{H}_2 + \text{Cl}_2 \rightarrow 2\text{HCl}$  is 182 KJ. Bond energies of H – H and Cl – Cl are 430 and 242 KJ/mol respectively. The H – Cl bond energy is:

- A** 245 kJ mol<sup>-1</sup>
- B** 427 kJ mol<sup>-1</sup>
- C** 336 kJ mol<sup>-1</sup>
- D** 154 kJ mol<sup>-1</sup>

## Question

If bond dissociation energies of  $\text{N} \equiv \text{N}$ ,  $\text{H} - \text{H}$  and  $\text{N} - \text{H}$  are  $x_1$ ,  $x_2$  and  $x_3$  respectively, hence enthalpy of formation of  $\text{NH}_3(\text{g})$  is.

- A**  $x_1 + 3x_2 - 6x_3$
- B**  $3x_3 - \frac{1}{2}x_1 - \frac{3}{2}x_2$
- C**  $\frac{x_1}{2} + \frac{3}{2}x_2 - 3x_3$
- D**  $6x_3 - x_1 - 3x_2$

## Question

The enthalpy change for the following reaction is 368 kJ. Calculate the average O–F bond energy



**A** 184 kJ/mol

**B** 368 kJ/mol

**C** 536 kJ/mol

**D** 736 kJ/mol

## Question

**AB, A<sub>2</sub> and B<sub>2</sub> are diatomic molecules. If the bond enthalpies of A<sub>2</sub>, AB and B<sub>2</sub> are in the ratio 1 : 1 : 0.5 and enthalpy of formation of AB from A<sub>2</sub> and B<sub>2</sub> is -100 kJ/mol<sup>-1</sup>. What is the bond enthalpy of A<sub>2</sub>.**

- A** 400 kJ/mol
- B** 200 kJ/mol
- C** 100 kJ/mol
- D** 300 kJ/mol

## Question

**Given below are two statements: One is labelled as Assertion (A) and the other is labelled as Reason (R)**

**Assertion (A): Enthalpy of neutralisation of strong monobasic acid with strong monoacidic base is always  $-57 \text{ kJ mol}^{-1}$**

**Reason (R): Enthalpy of neutralisation is the amount of heat liberated when one mole of  $\text{H}^+$  ions furnished by acid combine with one mole of  $\text{OH}^-$  ions furnished by base to form one mole of water.**

**In the light of the above statements, choose the correct answer from the options given below.**

- A** (A) is true but (R) is false
- B** Both (A) and (R) are true and (R) is the correct explanation of (A)
- C** (A) is false but (R) is true
- D** Both (A) and (R) are true but (R) is not the correct explanation of (A)

## Question

How much heat is liberated when 100 mL of 0.1 M NaOH are completely neutralised by 100 mL of 0.1 M HCl-

- A** -57 kJ
- B** -0.57 kJ
- C** -5.7 kJ
- D** -0.05 kJ

## Question

The enthalpy of neutralization of  $\text{NH}_4\text{OH}$  and  $\text{CH}_3\text{COOH}$  is  $-10.5$  kcal/mole and enthalpy of neutralization of strong base and  $\text{CH}_3\text{COOH}$  is  $-12.5$  kcal/mole. Calculate the enthalpy of dissociation of base  $\text{NH}_4\text{OH}$

**A** 3.0

**B** 4.0

**C** 2.0

**D** 10.0

## Question

Enthalpy of neutralisation of  $\text{CH}_3\text{COOH}$  by  $\text{NaOH}$  is  $-50.6 \text{ kJ/mol}$  and the heat of neutralisation of a strong acid with  $\text{NaOH}$  is  $-55.9 \text{ kJ/mol}$ . The value of  $\Delta H$  for the ionisation of  $\text{CH}_3\text{COOH}$  is

- A**  $3.5 \text{ kJ/mol}$
- B**  $4.6 \text{ kJ/mol}$
- C**  $5.3 \text{ kJ/mol}$
- D**  $6.4 \text{ kJ/mol}$

## Question

The enthalpy change for the conversion  $\frac{1}{2}\text{Cl}_2(\text{g})$  to  $\text{Cl}^-(\text{aq.})$  is (-) \_\_\_\_\_  $\text{kJmol}^{-1}$  (Nearest integer)

$$\text{Given, } \Delta_{\text{dis}} H_{\text{Cl}_2(\text{g})}^{\circ} = 240 \text{ kJmol}^{-1}$$

$$\Delta_{\text{eg}} H_{\text{Cl}_\text{g}}^{\circ} = -350 \text{ kJmol}^{-1}$$

$$\Delta_{\text{hyd}^-} H_{\text{Cl}^-(\text{g})}^{\circ} = -380 \text{ kJmol}^{-1}$$

## Question

The heat of solution of anhydrous  $\text{CuSO}_4$  and  $\text{CuSO}_4 \cdot 5\text{H}_2\text{O}$  are  $-70 \text{ kJ mol}^{-1}$  and  $+12 \text{ kJ mol}^{-1}$  respectively. The heat of hydration of  $\text{CuSO}_4$  to  $\text{CuSO}_4 \cdot 5\text{H}_2\text{O}$  is  $-x \text{ kJ}$ . The value of  $x$  is\_\_\_\_\_.

## Question

**Consider the given data:**

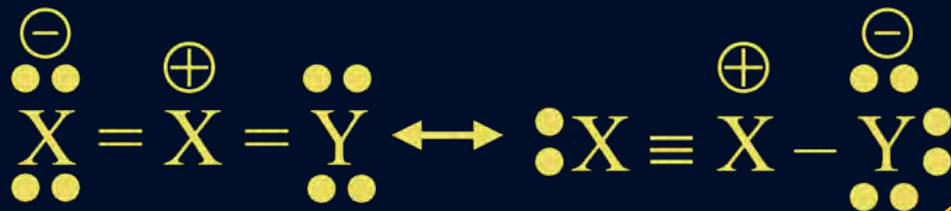


**Choose the correct statement:**

- A** Dissolution of gas in water is an endothermic process
- B** The heat of solution depends on the amount of solvent.
- C** The heat of dilution for the  $\text{HCl}(\text{HCl} \cdot 10\text{H}_2\text{O} \text{ to } \text{HCl} \cdot 40\text{H}_2\text{O})$  is  $3.78 \text{ kJ mol}^{-1}$ .
- D** The heat of formation of  $\text{HCl}$  solution is represented by both (A) and (B)

## Question

Resonance in  $X_2Y$  can be represented as



The enthalpy of formation of  $X_2Y$   $\left( X \equiv X(g) + \frac{1}{2} Y = Y(g) \rightarrow X_2Y(g) \right)$  is  $80 \text{ kJ mol}^{-1}$ . The magnitude of resonance energy of  $X_2Y$  is \_\_\_\_\_  $\text{kJ mol}^{-1}$  (nearest integer value)

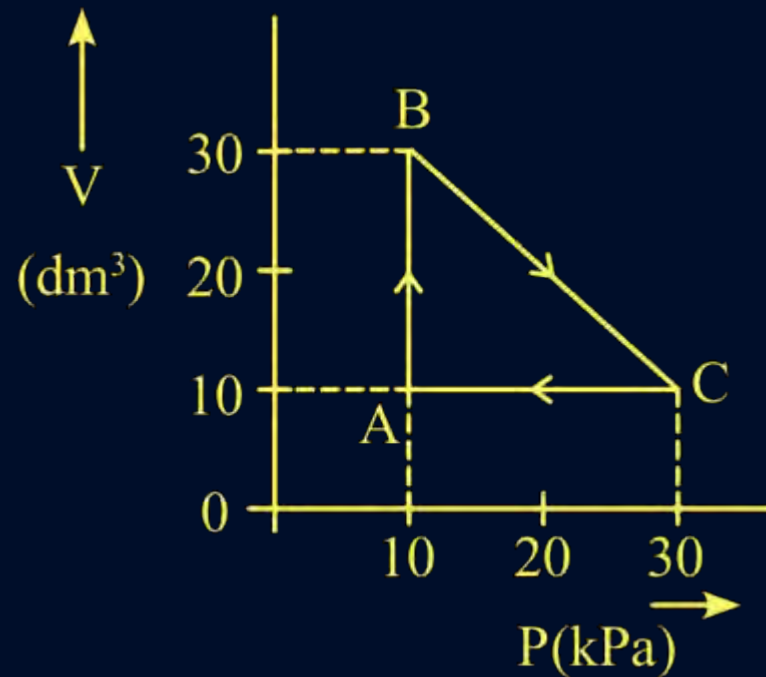
Given : Bond energies of  $X \equiv X$ ,  $X = X$ ,  $Y = Y$  and  $X = Y$  are 940, 410, 500 and  $602 \text{ kJ mol}^{-1}$  respectively.

valence X: 3, Y: 2

## Question

If 5 moles of an ideal gas expands from 10 L to a volume of 100 L at 300 K under isothermal and reversible condition then work,  $w$ , is  $-x$  J. The value of  $x$  is \_\_\_\_\_  
(Given  $R = 8.314 \text{ J K}^{-1} \text{ mol}^{-1}$ )

## Question



An ideal gas undergoes a cyclic transformation starting from the point A and coming back to the same point by tracing the path  $A \rightarrow B \rightarrow C \rightarrow A$  as shown in the diagram. The total work done in the process is \_\_\_\_ J.

## Question

Choose the correct option for free expansion of an ideal gas under adiabatic condition from the following:

**A**  $q = 0, \Delta T \neq 0, w = 0$

**B**  $q = 0, \Delta T < 0, w \neq 0$

**C**  $q \neq 0, \Delta T = 0, w = 0$

**D**  $q = 0, \Delta T = 0, w = 0$

## Question

An ideal gas,  $\bar{C}_V = \frac{5}{2}R$ , is expanded adiabatically against a constant pressure of 1 atm until it doubles in volume. If the initial temperature and pressure is 298 K and 5 atm, respectively, then the final temperature is \_\_\_\_\_K (nearest integer).

[  $\bar{C}_V$  is the molar heat capacity at constant volume ]

## Question

1 mole of ideal gas is allowed to expand reversibly and adiabatically from a temperature of  $27^{\circ}\text{C}$ . The work done is  $3 \text{ kJ mol}^{-1}$ . The final temperature of the gas is \_\_\_\_\_ K (Nearest integer). Given  $C_v = 20 \text{ J mol}^{-1}\text{K}^{-1}$ .

## Question

Let us consider a reversible reaction at temperature,  $T$ . In this reaction, both  $\Delta H$  and  $\Delta S$  were observed to have positive values. If the equilibrium temperature is  $T_e$ , then the reaction becomes spontaneous at:

- A**  $T = T_e$
- B**  $T_e > T$
- C**  $T > T_e$
- D**  $T_e = 5T$

## Question

**Match List-I with List-II.**

**Choose the correct answer from the options given below:**

**A** A-ii, B-i, C-iv, D-iii

**B** A-ii, B-iii, C-i, D-iv

**C** A-ii, B-i, C-iii, D-iv

**D** A-i, B-ii, C-iv, D-iii

| List-I<br>(Partial Derivatives) |                                                | List-II<br>(Thermodynamic Quantity) |       |
|---------------------------------|------------------------------------------------|-------------------------------------|-------|
| A.                              | $\left(\frac{\partial G}{\partial T}\right)_P$ | (i)                                 | $C_p$ |
| B.                              | $\left(\frac{\partial H}{\partial T}\right)_P$ | (ii)                                | $-S$  |
| C.                              | $\left(\frac{\partial G}{\partial P}\right)_T$ | (iii)                               | $C_v$ |
| D.                              | $\left(\frac{\partial U}{\partial T}\right)_V$ | (iv)                                | $V$   |

## Question

Ice at  $-5^{\circ}\text{C}$  is heated to become vapor with temperature of  $110^{\circ}\text{C}$  at atmospheric pressure. The entropy change associated with this process can be obtained from

**A** 
$$\int_{268\text{K}}^{273\text{K}} C_{p,m} dT + \frac{\Delta H_{m, \text{fusion}}}{T_f} + \frac{\Delta H_{m, \text{vaporisation}}}{T_b}$$
$$+ \int_{273\text{K}}^{373\text{K}} C_{p,m} dT + \int_{373\text{K}}^{383\text{K}} C_{p,m} dT$$

**B** 
$$\int_{268\text{K}}^{383\text{K}} C_p dT + \frac{\Delta H_{\text{melting}}}{273} + \frac{\Delta H_{\text{boiling}}}{373}$$

**C** 
$$\int_{268\text{K}}^{273\text{K}} \frac{C_{p,m}}{T} dT + \frac{\Delta H_{m, \text{fusion}}}{T_f} + \frac{\Delta H_{m, \text{vaporisation}}}{T_b} +$$
$$\int_{273\text{K}}^{373\text{K}} \frac{C_{p,m} dT}{T} + \int_{373\text{K}}^{383\text{K}} \frac{C_{p,m} dT}{T}$$

**D** 
$$\int_{268\text{K}}^{383\text{K}} C_p dT + \frac{q_{\text{rev}}}{T}$$

## Question

Which of the following is not correct?

- A**  $\Delta G$  is negative for a spontaneous reaction
- B**  $\Delta G$  is positive for a spontaneous reaction
- C**  $\Delta G$  is zero for a reversible reaction
- D**  $\Delta G$  is positive for a non-spontaneous reaction

## Question

Which of the following relations are correct?

(A)  $\Delta U = q + P\Delta V$

(B)  $\Delta G = \Delta H - T\Delta S$

(C)  $\Delta S = \frac{q_{\text{rev}}}{T}$

(D)  $\Delta H = \Delta U - \Delta nRT$

Choose the most appropriate answer from the options given below:

**A** (C) and (D) only

**B** (B) and (C) only

**C** (A) and (B) only

**D** (B) and (D) only

## Question

**Match the List-I with List-II**

**Choose the correct answer from the option given below:**

**A** (A)-(III), (B)-(II), (C)-(IV), (D)-(I)

**B** (A)-(II), (B)-(III), (C)-(IV), (D)-(I)

**C** (A)-(II), (B)-(III), (C)-(I), (D)-(IV)

**D** (A)-(II), (B)-(I), (C)-(III), (D)-(IV)

| List-I |                                              | List-II |                                                                                  |
|--------|----------------------------------------------|---------|----------------------------------------------------------------------------------|
| A.     | Spontaneous process                          | I.      | $\Delta H < 0$                                                                   |
| B.     | Process with $\Delta P = 0$ , $\Delta T = 0$ | II.     | $\Delta G_{T,P} < 0$                                                             |
| C.     | $\Delta H_{\text{reaction}}$                 | III.    | Isothermal and Isobaric process                                                  |
| D.     | Exothermic Process                           | IV.     | [Bond energies of molecules in reactants] - [Bond energies of product molecules] |

## Question

The incorrect expression among the following is:

**A**  $\ln K = \frac{\Delta H^\circ - T\Delta S^\circ}{RT}$

**B** For isothermal process,  $w_{\text{reversible}} = -nRT \ln \frac{V_f}{V_i}$

**C**  $\frac{\Delta G_{\text{System}}}{\Delta S_{\text{Total}}} = -T$  ( at constant P)

**D**  $K = e^{-\Delta G^\circ/RT}$

## Question

**During which of the following processes, does entropy decrease?**

**(A) Freezing of water to ice at  $0^{\circ}\text{C}$**

**(B) Freezing of water to ice at  $-10^{\circ}\text{C}$**

**(C)  $\text{N}_2(\text{g}) + 3\text{H}_2(\text{g}) \rightarrow 2\text{NH}_3(\text{g})$**

**(D) Adsorption of  $\text{CO}(\text{g})$  on lead surface**

**(E) Dissolution of  $\text{NaCl}$  in water**

**Choose the correct answer from the options given below:**

**A** (A) and (E) only

**B** (A), (B), (C) and (D) only

**C** (A), (C) and (E) only

**D** (B) and (C) only

## Question

One mole of an ideal gas at 350K is in a 2.0 L vessel of thermally conducting walls, which are in contact with the surroundings. It undergoes isothermal reversible expansion from 2.0L to 3.0L against a constant pressure of 4 atm. The change in entropy of the surroundings ( $\Delta S$ ) is \_\_\_\_\_ J K<sup>-1</sup> (Nearest integer)  
Given:  $R = 8.314 \text{ J K}^{-1} \text{ Mol}^{-1}$ .

## Question

Data given for the following reaction is as follows:



| Substance   | $\Delta_f H^\circ (\text{kJ mol}^{-1})$ | $\Delta S^\circ (\text{J mol}^{-1} \text{K}^{-1})$ |
|-------------|-----------------------------------------|----------------------------------------------------|
| FeO(s)      | -266.3                                  | 57.49                                              |
| C(graphite) | 0                                       | 5.74                                               |
| Fe(s)       | 0                                       | 27.28                                              |
| CO(g)       | -110.5                                  | 197.6                                              |

The minimum temperature in K at which the reaction becomes spontaneous is  
\_\_\_ (Integer answer)

## Question

**The correct statement amongst the following is:**

- A** The term 'standard state' implies that the temperature is  $0^{\circ}\text{C}$
- B** The standard state of a pure gas is the pure gas at a pressure of 1 bar and temperature 273 K
- C**  $\Delta_{\text{f}}H^{\theta}_{298}$  is zero for  $\text{O}(\text{g})$
- D**  $\Delta_{\text{f}}H^{\theta}_{500}$  is zero for  $\text{O}_2(\text{g})$

## Question

**The true statement amongst the following is:**

- A**  $S$  is a function of temperature but  $\Delta S$  is not a function of temperature.
- B** Both  $\Delta S$  and  $S$  are functions of temperature.
- C**  $S$  is not a function of temperature but  $\Delta S$  is a function of temperature.
- D** Both  $S$  and  $\Delta S$  are not functions of temperature

## Question

The number of endothermic process/es from the following is \_\_\_\_\_

- A**  $\text{I}_2(\text{g}) \rightarrow 2\text{I}(\text{g})$
- B**  $\text{HCl}(\text{g}) \rightarrow \text{H}(\text{g}) + \text{Cl}(\text{g})$
- C**  $\text{H}_2\text{O}(\text{l}) \rightarrow \text{H}_2\text{O}(\text{g})$
- D**  $\text{C}(\text{s}) + \text{O}_2(\text{g}) \rightarrow \text{CO}_2(\text{g})$
- E** Dissolution of ammonium chloride in water

## Question

For a chemical reaction  $A + B \rightleftharpoons C + D$

( $\Delta_r H^\circ = 80 \text{ kJ mol}^{-1}$ ) the entropy change  $\Delta_r S^\circ$  depends on the temperature  $T$  (in K) as  $\Delta_r S^\circ = 2T \text{ (JK}^{-1} \text{ mol}^{-1}\text{)}$ .

Minimum temperature at which it will become spontaneous is \_\_\_\_\_ K  
(Integer).

## Question

The standard reaction Gibbs energy for a chemical reaction at an absolute temperature  $T$  is given by:

$$\Delta_r G^\circ = A - BT$$

Where  $A$  and  $B$  are non-zero constant. Which of the following is TRUE about this reaction?

- A** Endothermic if  $A > 0$
- B** Exothermic if  $A > 0$  and  $B < 0$
- C** Endothermic if  $A < 0$  and  $B > 0$
- D** Exothermic if  $B < 0$

## Question

For the chemical reaction  $X \rightleftharpoons Y$ , the standard reaction Gibbs energy depends on temperature  $T$  (in K) as

$$\Delta_r G^\circ \text{ (in kJ mol}^{-1}\text{)} = 120 - \frac{3}{8} T$$

The major component of the reaction mixture at  $T$  is:

- A** Y if  $T = 300$  K
- B** Y if  $T = 280$  K
- C** X if  $T = 350$  K
- D** X if  $T = 315$  K